Output:-

Que :- A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Optimal page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3)

Code:-

```
GNU nano 7.2
#include <stdio.h>

int main() {
    int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};
    int n = 12, frames = 3;
    int frame[3];
    int i, j, k, pos, hits = 0, faults = 0;

    for(i = 0; i < frames; i++)
        frame[i] = -1;

    printf("Page Reference String: ");
    for(i = 0; i < n; i++)
        printf("%d ", pages[i]);

    printf("\nNumber of Frames: %d\n\n", frames);

    for(i = 0; i < n; i++) {
        int found = 0;

        for(j = 0; j < frames; j++) {
            if(frame[j] == pages[i]) {
                hits++;
                found = 1;
                break;
            }
        }

        if(!found) {
            int farthest = i, index = -1;

            for(j = 0; j < frames; j++) {
                for(k = i+1; k < n; k++) {
                    if(frame[j] == pages[k])
                        break;
                }

                if(k == n) {
                    index = j;
                    break;
                }
```

```
GNU nano 7.2
                found = 1;
                break;
            }
        }

        if(!found) {
            int farthest = i, index = -1;

            for(j = 0; j < frames; j++) {
                for(k = i+1; k < n; k++) {
                    if(frame[j] == pages[k])
                        break;
                }

                if(k == n) {
                    index = j;
                    break;
                }

                if(k > farthest) {
                    farthest = k;
                    index = j;
                }
            }

            if(index == -1)
                index = 0;

            frame[index] = pages[i];
            faults++;
        }
    }

    printf("\nPage Hits = %d\n", hits);
    printf("Page Faults = %d\n", faults);
    printf("Hit Ratio = %.2f%%\n", (hits/(float)n)*100);
    printf("Fault Ratio = %.2f%%\n", (faults/(float)n)*100);

    return 0;
```

Output:-

```
himani@DELL:~$ nano optimal.c
himani@DELL:~$ gcc optimal.c -o optimal./optimal
/usr/bin/ld: cannot open output file optimal./optimal: No such file or directory
collect2: error: ld returned 1 exit status
himani@DELL:~$ gcc optimal.c -o optimal
himani@DELL:~$ ./optimal
Page Reference String: 1 2 3 4 1 2 5 1 2 3 4 5
Number of Frames: 3


Page Hits = 5
Page Faults = 7
Hit Ratio = 41.67%
Fault Ratio = 58.33%
himani@DELL:~$
```